

Design and Analysis of Algorithms (DAA) - Lab 7 Assignment

Greedy Techniques & Graph Algorithms: Minimum Spanning Trees

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GitHub Repository: <https://github.com/kathichetanreddy8106-ui/DESIGN-AND-ANALYSIS-OF-ALGORITHMS>

1. Minimum Spanning Tree Implementation using Kruskal's Algorithm

SOURCE CODE

```
#include <stdio.h>
int p[10];
int find(int x) {
    while(p[x] != x)
        x=p[x];
    return x;
}
int main() {
    int n, e, a[20][3], i, j, k, t, c=0, cost=0, x, y;
    scanf("%d %d", &n, &e);
    for(i=0; i<e; i++)
        scanf("%d %d %d", &a[i][0], &a[i][1], &a[i][2]);
    for(i=0; i<n; i++) p[i]=i;
    for(i=0; i<e-1; i++)
        for(j=i+1; j<e; j++)
            if(a[i][2] > a[j][2])
                for(k=0; k<3; k++) t=a[i][k], a[i][k]=a[j][k],
                    a[j][k]=t;
    for(i=0; i<e && c<n-1; i++) {
        x=find(a[i][0]); y=find(a[i][1]);
        if(x != y) {
            p[x]=y;
            printf("%d - %d = %d\n", a[i][0], a[i][1], a[i][2]);
            cost += a[i][2]; c++;
        }
    }
    printf("cost = %d", cost);
}
```

PROGRAM OUTPUT

```
5 6 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 1 - 2 = 3 4 - 5 = 6 10 - 11 = 12 cost = 21
```

COMPILER EXECUTION SCREENSHOT

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main.c | Run

```
1 #include <stdio.h>
2 int p[10];
3 int find(int x) {
4     while(p[x] != x)
5         x=p[x];
6     return x; }
7 int main() {
8     int n,e,a[20][3],i,j,k,t,c=0,cost=0,x,y;
9     scanf("%d %d",&n,&e);
10    for(i=0; i<e; i++)
11        scanf("%d %d %d",&a[i][0],&a[i][1],&a[i][2]);
12    for(i=0; i<n; i++) p[i]=i;
13    for(i=0; i<e-1; i++)
14        for(j=i+1; j<e; j++)
15            if(a[i][2] > a[j][2])
16                for(k=0; k<3; k++) t=a[i][k], a[i][k]=a[j][k],
17                    a[j][k]=t;
18    for(i=0; i<e && c<n-1; i++) {
```

Output

```
5 6
1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
1 - 2 = 3
4 - 5 = 6
cost = 9
```

Type here to search | Rain coming | ENG IN | 7:39 PM | 9/21/2026

Programiz Online C Compiler Execution for Kruskal's Algorithm

2. Minimum Spanning Tree Implementation using Prim's Algorithm

SOURCE CODE

```
#include <stdio.h>
#define I 999
int main() {
    int n, a[10][10], v[10]=0, i, j, k, min=I, cost=0, u, y;
    scanf("%d", &n);
    for(i=0; i<n; i++)
        for(j=0; j<n; j++)
            scanf("%d", &a[i][j]);
    v[0]=1;
    for(k=0; k<n-1; k++) {
        min = I;
        for(i=0; i<n; i++)
            if(v[i])
                for(j=0; j<n; j++)
                    if(!v[j] && a[i][j] < min)
                        min = a[i][j], u=i, y=j;
        v[y] = 1;
        printf("%d - %d = %d\n", u, y, min);
        cost += min;
    }
    printf("cost = %d", cost);
}
```

PROGRAM OUTPUT

```
3 1 2 3 4 5 6 7 8 9 0 - 1 = 2 0 - 2 = 3 cost = 5
```

COMPILER EXECUTION SCREENSHOT

The screenshot shows the Programiz Online C Compiler interface. The browser address bar displays `programiz.com/c-programming/online-compiler/`. The page header includes the Programiz logo, the text "C Online Compiler", and a "Programiz PRO" button. A banner at the top reads "Stop copy pasting code you don't actually understand" with the subtext "Build the coding confidence you need to become a developer companies will fight for" and a "Become a PRO" button.

The main editor area shows a file named `main.c` with the following C code:

```
1 #include <stdio.h>
2 #define I 999
3 int main() {
4     int n,a[10][10],v[10]={0},i,j,k,min=I,cost=0,u,y;
5     scanf("%d",&n);
6     for(i=0; i<n; i++)
7         for(j=0; j<n; j++)
8             scanf("%d",&a[i][j]);    v[0]=1;
9     for(k=0; k<n-1; k++) {
10         min = I;
11         for(i=0; i<n; i++)
12             if(v[i])
13                 for(j=0; j<n; j++)
14                     if(!v[j] && a[i][j] < min)
15                         min = a[i][j], u=i, y=j;
16         v[y] = 1;
17         printf("%d - %d = %d\n",u,y,min);
18         cost += min;
19     }
```

The "Output" panel on the right shows the execution results:

```
3
1 2 3 4 5 6 7 8 9
0 - 1 = 2
0 - 2 = 3
cost = 5
```

The Windows taskbar at the bottom shows the system clock as 7:41 PM on 9/21/2026, with the language set to ENG IN.

Programiz Online C Compiler Execution for Prim's Algorithm